Oct 13, 2026 (Due: 08:00 Oct 20, 2026)

1. Let $A \in \mathbb{C}^{m\times n}$ and $X \in \mathbb{C}^{n\times m}$. Suppose that for any $b \in \mathbb{C}^m$, $x = Xb$ is always a minimizer of the least squares problem $\min_x \|Ax - b\|_2$. Show that $AXA = A$ and $(AX)^* = AX$.

2. Given $A \in \mathbb{C}^{n\times n}$, $r \in \mathbb{C}^n$. Design an algorithm to construct an orthonormal basis of the *Krylov subspace*

$$\mathcal{K}(A, r, k) = \text{span}\left\{r, Ar, A^2r, \ldots, A^{k-1}r\right\}.$$

Implement your algorithm. Test your implementation with a few examples and visualize the loss of orthogonality.

3. Let A be a square matrix, whose eigenvalues are λ_1, λ_2, ..., λ_n, ordered with descending magnitudes. Suppose that $\lambda_1 = \lambda_2 = \cdots = \lambda_k$ and $|\lambda_k| > |\lambda_{k+1}|$. Discuss the behavior of the power iteration applied to this matrix.

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$$A = \begin{bmatrix} \lambda & 1 \\ 0 & -\lambda \end{bmatrix},$$

where $\lambda \in \mathbb{C}$ is a given nonzero constant.

5. Randomly generate a 1000×1000 matrix A with positive entries. Use the power iteration to compute $\rho(A)$. Visualize the convergence history.

6. (H) Inplement the mixed-precision algorithm for QRCP discussed in the lecture. Compare the performance of the mixed-precision algorithm with the fixed-precision one.

7. (optional) Let $A \in \mathbb{R}^{m\times n}$ with full column rank. Establish the connection between the Householder-QR algorithm applied to the matrix $[0, A^\top]^\top \in \mathbb{R}^{(m+n)\times n}$ and the MGS algorithm applied to A.

8. (optional) Given a vector $d \in \mathbb{C}^n$ and two matrices $W \in \mathbb{C}^{n\times m}$, $V \in \mathbb{C}^{n\times p}$, with $m \le n \le m + p$. Suppose that

$$\text{rank}([W, V]) = n, \qquad \text{rank}(W) = m.$$

Design an algorithm to solve the *generalized least squares (GLS) problem*

$$\min_{x\in\mathbb{C}^m,\ y\in\mathbb{C}^p} \|y\|_2 \quad \text{s.t.} \quad Wx + Vy = d.$$

(H, optional) Design an iterative refine algorithm to solve the GLS problem.

9. (H, optional) Design a level-3 variant of the GTH algorithm.